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Aqueous biphasic systems developed with deep eutectic solvents and polymer for the efficient extraction of pigments from beverages

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ABSTRACT

Novel aqueous biphasic systems (ABSs) developed with benzyl-based quaternary ammonium salts-deep eutectic solvents (DESs) and polypropylene glycol (PPG) were herein proposed. The liquid-liquid equilibrium and the partitioning behavior of pigments in the systems were addressed. The results suggested that the shorter the carbon chain length of the DES, the easier to form two phases. The analysis of mixed samples showed that the selective separation was achieved in the ABSs, including 99.47% of tartrazine in the DES-rich phase and 98.47% of sudan III in the PPG-rich phase. Additionally, the systems were successfully applied to the extraction of pigments from the actual beverage samples with recoveries ranging from 93.43% to 102.15%. Furthermore, the study on the separation mechanism indicated that the hydrogen bonding played a significant role in the separation process. All the above results highlight the proposed DES/polymer-based ABSs have great advantages in selective and high-performance separation of pigments from beverages.

1. Introduction

Owing to simple operation, moderate conditions and scalable potential, aqueous biphasic systems (ABSs) are considered as a promising technology in the field of extraction and separation. Generally, these systems can be formed by two distinct water-soluble components which are dissolved in water until their concentration exceeds a critical value (Chao & Shum, 2020; Mendes, Rosa, Ramalho, Freire, & e Silva, F. A., 2023). After two phases are generated, there are two independent environments with different physical and chemical properties (Chen et al., 2019; Chen et al., 2021; Lyu, Meany, Yang, Li, & Zheng, 2019). Once analytes are introduced into these systems, the choice between two phases is largely dependent on solubility parameters (Pereira, Rebelo, Rogers, Coutinho, & Freire, 2013). Most of analytes tend to distribute into the phases with similar solubility parameters, which provide a guideline for reasonably designing the systems. Various constituents have been applied into the design of ABSs, such as two kinds of inorganic salt solution (Bridges, Gutowski, & Rogers, 2007) and two types of polymers (Jorge, Silva, Coutinho, & Pereira, 2024; Khripin, Fagan, & Zheng, 2013). However, the limited polarity between the two phases of the conventional ABSs has restricted their further applications in separation process.

Deep eutectic solvent (DES), as a new component of ABSs, has evoked enormous attention due to easy preparation, low cost, devisable structures and effective performance (Cai, Ma, Li, & Tan, 2024; Deng, Zong, & Xiao, 2017; Xu et al., 2018; Zeng et al., 2014). By heating hydrogen bond donor (HBD) and hydrogen bond acceptor (HBA) with a proper molar ratio, DES can be obtained at a certain temperature (Abbott, Boothby, Capper, Davies, & Rasheed, 2004; Abbott, Capper, Davies, Rasheed, & Tambyrajah, 2003; Francisco, van den Bruinhorst, & Kroon, 2013). Up to now, the study on DES-based ABSs has focused on exploiting new phase-forming components, improving extraction efficiency and selective separation. For example, polypropylene glycol 400 (PPG 400)-based DES was prepared and combined with new types of salts such as glycine, betaine, choline chloride (ChCl), sucrose, sorbitol and glucose to construct novel ABSs (Zhang et al., 2018). After optimizing the conditions, the extraction efficiency of the pigment could reach 98.00% and the selective partition of pigments was achieved. ABSs composed of two DESs, including hexafluoroisopropanol (HFIP)-based DES and glycerol-based DES, were also established and applied to address the selective separation of dyes (Xu, Xu, & Wang, 2020). Besides, the partition of astaxanthin in fructose-based DES/ionic liquid (IL)

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ABSs was investigated (Zhang, Li, & Gao, 2024). The results indicated that the astaxanthin ester and the free astaxanthin could be selectively separated with a coefficient specificity of 48.31. However, there are still some haunting obstacles, such as despite there are multiple combinations of HBA and HBD to prepare DESs, the HBA adopted in this area are restricted to common aliphatic quaternary ammonium salts, for example, ChCl, betaine and tetramethylammonium chloride. In addition, the abovementioned new-types of salts suffered from the drawbacks of relatively weak phase-forming ability, the preparation of two DESs or DES and IL were also complicated. Therefore, it is intriguing to synthesize novel DES and construct new ABSs to provide more enhanced platform for extraction and separation.

Benzyl-based quaternary ammonium salts are aromatic compounds and their hydrophilicity can be easily manipulated by altering the carbon chain length and anion species (Priyanka, Basaiahgari, & Gardas, 2017). It is worth trying to probe the potential of aromatic quaternary ammonium salts as HBA in forming ABSs. Meanwhile, the DES-based ABSs have been extended up to DES/polymer combinations (Baghlani & Sadeghi, 2018; Marchel, Coroadinha, & Marrucho, 2020; Ahsaie & Pazuki, 2021; Lazarević, Mušović, Trtić-Petrović, & Gadžurić, 2023). PPG is commonly used as the phase-forming polymer due to its excellent phase-forming ability, low cost, high biodegradability and low volatility (Zafarani-Moattar, Hamzehzadeh, & Nasiri, 2012). Thus, the benzyl-based quaternary ammonium salts-DES is combined with PPG to construct novel ABSs, aiming at further extending the hydrophobicity-hydrophilicity range and improving the extraction performance of DES-based ABSs.

Pigments are diverse in polarities, which makes them can be used as a prober to demonstrate the relative hydrophobicity of the upper and the lower phases of the ABSs. Besides, the pigments have been widely used as food additives to provide color to foods. However, the overuse of pigments in some case represents a potential risk to human health (Authority, 2010; Silva, Reboredo, & Lidon, 2022). Thus, the sensitive and accurate determination of pigments content in food is of great necessity (Peksa et al., 2015). Considering the preconcentration of pigments from the complex matrix is a crucial step before pigment detection, it is significant to develop a novel approach with high performance to separate and enrich pigments.

Herein, four kinds of benzyl-based quaternary ammonium salts, including benzyltrimethylammonium chloride (BTMAC), benzyltrimethylammonium bromide (BTMAB), benzyltriethylammonium chloride (BTEAC) and benzyltriethylamine bromide (BTEAB), were chosen as the HBAs. Glycerol was selected as the HBD for its low cost and biocompatibility. PPG with molecular weights of 400, 600 and 1000 were used as the phase-forming polymers. The phase behaviors of the DES/polymer-based ABSs were determined. Four types of pigments with different hydrophobicities, including amaranth (AM), tartrazine (TA), sunset yellow FCF (SY) and sudan III (SU), were adopted as the model analytes. The influence of the hydrophobicities of the phase components on pigment distribution was investigated. In addition, as a contrast, a conventional ABS consisting of DES and salt was constructed. Both the two DES-based ABSs were explored in the influence factors. And the extraction experiments of mixed samples were also carried out to evaluate the selective extraction ability of the two systems. After that, the separation of actual beverage samples was performed to verify the practical application of the DES/polymer-based ABSs. Finally, combined with UV–vis spectra, FT-IR spectra, the influence of solution pH on pigment partition and dynamic light scattering (DLS), the extraction mechanism was discussed.

2. Materials and methods

2.1. Reagents

All the chemicals used were analytical grades. Benzyltrimethylammonium chloride (BTMAC), benzyltrimethylammonium bromide (BTMAB), benzyltriethylammonium chloride (BTEAC) and benzyltriethylamine bromide (BTEAB) were purchased from Aladdin Reagent Co., Ltd. (Shanghai, China). Amaranth (AM), tartrazine (TA), sunset yellow FCF (SY) and sudan III (SU) (as shown in Table S1) were gotten from Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China). Glycerin, polypropylene glycol 400 (PPG 400), polypropylene glycol 600 (PPG 600) and polypropylene glycol 1000 (PPG 1000) were bought from Adamas Reagent Co., Ltd. (Shanghai, China). Dipotassium phosphate and monopotassium phosphate were obtained from Titan Technology Co., Ltd. (Shanghai, China).

2.2. Preparation of DESs

In this work, benzyltrimethylammonium chloride (BTMAC), benzyltrimethylammonium bromide (BTMAB), benzyltriethylammonium chloride (BTEAC) and benzyltriethylammonium bromide (BTEAB) were chosen as the hydrogen bond acceptors (HBAs). Glycerin (Gly) was selected as the hydrogen bond donor (HBD). Four kinds of DESs were synthesized by mixing HBA and HBD in molar ratio of 1:4 (as illustrated in Table S2). The mixture was stirred for 1 h with the temperature maintained at 80 °C. And the synthetic process of DES was presented in Fig. S1 by taking [BTMAC][Gly] as an example.

2.3. Preparation of phase diagrams

The binodal curves were determined by the cloud point titration method at ambient temperature and atmospheric pressure (Gutowski et al., 2003). A certain amount of DES aqueous solutions with a mass fraction of about 80% were weighed and added to a 15 mL centrifuge tube. Then, pure PPG was added drop by drop until the solution became cloudy, indicative of forming biphasic region. Then, a known amount of ultrapure water was added to the mixture until the solution turned clear, demonstrating the monophasic region was achieved. Significantly, after each addition of PPG and ultrapure water, the solution was shaken in a thermostatic oscillator at 200 rpm and then centrifuged at 2000 rpm for 7 min to ensure the phase splitting. The above steps were repeated to attain enough data to plot the phase diagram, and the curves were determined by weight quantification within $\pm 10^{-4}$ g.

2.4. Construction of DES-based ABSs

Two types of ABSs were constructed in our study, including DES/PPG-based ABSs and DES/salt-based ABSs. Briefly, 1.5 g DESs ([BTMAC][Gly], [BTMAB][Gly], [BTEAC][Gly] and [BTEAB][Gly]), 1.6 g PPG (PPG 400, PPG 600, PPG 1000) and 0.8 mL water were added into centrifuge tubes, and the systems were placed in a thermostatic oscillator to shake for 20 min at 200 rpm. Then the centrifuges were centrifuged at 2000 rpm for 7 min to constitute DES/PPG-based ABSs. The upper phase and the bottom phase in the ABSs corresponded to PPG-rich phase and DES-rich phase, respectively. The DES/salt-based ABS consisted of 1.0 g [BTMAC][Gly], 1.7 g K_2HPO_4 and 2 mL water. The upper phase and the bottom phase mainly consisted of DES and salt, respectively.

2.5. The stability of DES in ABSs

To evaluate the DES stability in the DES/PPG-based ABSs, 1.5 g [BTMAC][Gly] with molar ratio of 1:4, 1.6 g PPG 1000 and 0.8 mL water were used to construct ABS. The systems were placed in a thermostatic oscillator to shake for 20 min at 200 rpm and then were centrifuged at 2000 rpm for 7 min to guarantee a complete phase-splitting. Afterwards, the top phase and the bottom phase were separated and characterized by ^{1}H NMR to explore the molar ratio of HBA and HBD. Besides, the water content in both phases was measured through a gravimetric method. The PPG-rich phase and the DES-rich phase were separated and dried in an air oven at 90 °C. The evaporated water can be calculated by subtracting the residue from the total mass.

2.6. Partitioning of pigments in ABSs

In order to investigate the extractability of the ABSs, four kinds of pigments with different properties, including AM, TA, SY and SU, were chosen as model analytes (as shown in Table S1). The extraction process was performed in a thermostatic oscillator and the speed was set at 200 rpm. Afterwards, the systems were centrifuged at 2000 rpm for 7 min to guarantee the absolute phase separation. Then the concentrations of pigments in two phases were determined by detecting the absorbance at 520 nm for amaranth, 426 nm for tartrazine, 480 nm for sunset yellow FCF and 502 nm for sudan III with a UV–vis spectrophotometer. The pigments were determined at a mixed solution when performing the experiment of real sample analysis. Otherwise, the pigments were determined separately in single standard solution. The samples were analyzed against the blanks containing the same constituents but without pigments to avoid interference from phase components. And each experiment was carried out three times and the results were recorded as the means ± standard deviations (SDs). The extraction yield (E) was obtained according to the following formula:

$$E = \frac{C_e V_e}{C_0 V_0} \times 100\% \tag{1}$$

where C_0 (mg•mL^{-1}) and V_0 (mL) are the initial concentration and volume of the pigment solution, respectively. Besides, in DES/PPG-based ABSs, C_e (mg•mL^{-1}) and V_e (mL) represent the equilibrium concentrations of pigments solution and the phase volumes in the bottom phase, respectively. As for DES/salt-based ABSs, C_e and V_e represent the equilibrium concentrations of pigments solution and the phase volumes in the top phase, respectively.

2.7. Sample preparation

Three kinds of beverages were obtained from a local supermarket, including carbonated drinks, energy drinks and juice. The carbonated drink and juice were diluted with ultrapure water for 2 times and 10 times, respectively. The energy drink was used directly without further dilution.

2.8. Apparatus

A DF-101S magnetic stirrer (Henan, China) was utilized to prepare DESs. The infrared spectrum was measured by an IRPrestige-21 FT-IR spectrometer (Shimadzu, Japan). ^{1}H NMR spectrum was recorded by a AV 500M NMR (Bruker, Germany). The extraction process was carried out in a SHZ-82 A incubator shaker (Nanjing, China). The concentrations of the pigments were detected by a UV-2600 UV–vis spectrophotometer (Shimadzu, Japan). A L600-A table-top centrifuge (Hunan, China) was used to ensure the phase splitting. A PHS-25 pH meter (Shanghai, China) was taken to prepare a series of phosphate buffered solutions. A ZSU3200 dynamic light scattering (Malvern, UK) was used to examine the size distribution of samples.

3. Results and discussion

3.1. Characterization of DESs

The structures of BTMAC, Gly and [BTMAC][Gly] were determined by FT-IR spectra as shown in Fig. S2. In the spectra of BTMAC, the peaks at 2881 cm^{-1} and 2937 cm^{-1} were assigned to the $\nu_{\text{C-H}}$ of CH_3 and CH_2, respectively. The peaks at 1458 cm^{-1} and 1651 cm^{-1} corresponded to the $\nu_{\text{C=C}}$ of a benzene ring. The peak located at 1218 cm^{-1} was caused by the $\nu_{\text{C-N}}$. In the spectra of Gly, the stretching vibrations at 1042 cm^{-1} and 1113 cm^{-1} belonged to the C—O of primary alcohol and secondary alcohol, respectively. It can be analyzed from the spectra of [BTMAC][Gly] that the absorption peaks of C—O in pure Gly at 1042 cm^{-1} and 1113 cm^{-1} shifted to 1041 cm^{-1} and 1110 cm^{-1}, respectively. The movement of the C—O stretching vibration towards the low wavenumber indicated the formation of hydrogen bonds between BTMAC and Gly. Besides, the absorption band within the range of 3500–3000 cm^{-1} was assigned to the tensile vibration of O—H. It can be detected that the absorption peak of O—H in [BTMAC][Gly] was stronger and wider compared to that of the pure materials, which further verified the existence of abundant hydrogen bonds in DES. The similar spectra were detected for the other three DESs, which were shown in Fig. S3.

The ^{1}H NMR was further adopted to characterize the chemical structures of the DESs. The spectra of BTMAC, Gly and [BTMAC][Gly] were collected and the spectra data was completely interpreted as illustrated in Fig. S4. Compared to the spectra of BTMAC and Gly, there was no new peaks appeared in that of [BTMAC][Gly], which meant no new substances and chemical bonds were formed in the synthesis of DES. Meanwhile, after the formation of DES, multiple peaks around 4.9 ppm in Gly were shifted to 4.7 ppm. Since the multiple peaks were ascribed to the -OH groups in Gly, it was inferred that the hydrogen bonds were formed between BTMAC and Gly (Haider, Jha, Kumar, & Sivagnanam, 2020). In addition, the spectra of [BTMAB][Gly], [BTEAC][Gly] and [BTEAB][Gly] were also determined as shown in Fig. S5, Fig. S6 and Fig. S7, respectively. From the integral values of the H element in these spectra, it can be calculated that the molar ratio of benzyl-based quaternary ammonium salts to Gly was about 1:4. As a result, the above analysis proved the successful preparation of DESs.

3.2. Equilibrium phase behavior

In this section, the phase diagrams of the DES/PPG-based ABSs were plotted to evaluate the phase-forming ability of the components. The system was divided into two phases when the mixtures of constituents were above the binodal curve. And the region below the curve indicated the system was homogeneous. The closer the curve was to the origin, the easier the system was to form two phases.

3.2.1. Effect of the types of DES

Four types of DESs were combined with PPG 1000 to construct ABSs to investigate the influence of the structures of DESs on the two-phase splitting. As illustrated in Fig. 1A, the following rank on the DESs potential to form two phases was detected: [BTMAB][Gly] > [BTMAC][Gly] > [BTEAB][Gly] > [BTEAC][Gly]. It was reported that the hydration competition between two phase-forming components was the reason for the formation of ABSs (Ferreira et al., 2016; Freire, Louros, Rebelo, & Coutinho, 2011; Pereira et al., 2013). In particular, the component with stronger hydration ability was the water structure-making, while the other one was the water structure-breaking (Wu, Wang, Pei, Wang, & Li, 2010). In the traditional DES/salt-based ABSs, the DES was salted-out by salt and it acted as a water structure-breaking component (Deng et al., 2017; Zhang et al., 2018). However, in the current work, the DESs were water structure-making components because of their stronger hydration ability than PPG. In other words, the PPG was salted-out by DESs and the DESs acted as salting-out agents. As a consequence, the ability of each DES to induce phase-splitting was related to its affinity for water (Pereira, Lima, Freire, & Coutinho, 2010). The higher the affinity for water, the stronger the phase-forming ability of DES. The hydrophilicity of molecules could be assessed by the logarithm of octanol/water partition coefficient (log K_{ow}). The log K_{ow} of BTMAC was −2.17, indicating its strong hydrophilicity. The increase in the alkyl chain length of DES would lead to a decrease of its hydrophilicity (Chen et al., 2021). As a result, the log K_{ow} of BTEAC increased to 1.39. The [BTMAC][Gly] showed a higher affinity for water than [BTEAC][Gly], which resulted in the stronger phase-forming ability of [BTMAC][Gly] compared to that of [BTEAC][Gly]. Meanwhile, the same trend was observed in [BTMAB][Gly] and [BTEAB][Gly]. When it comes

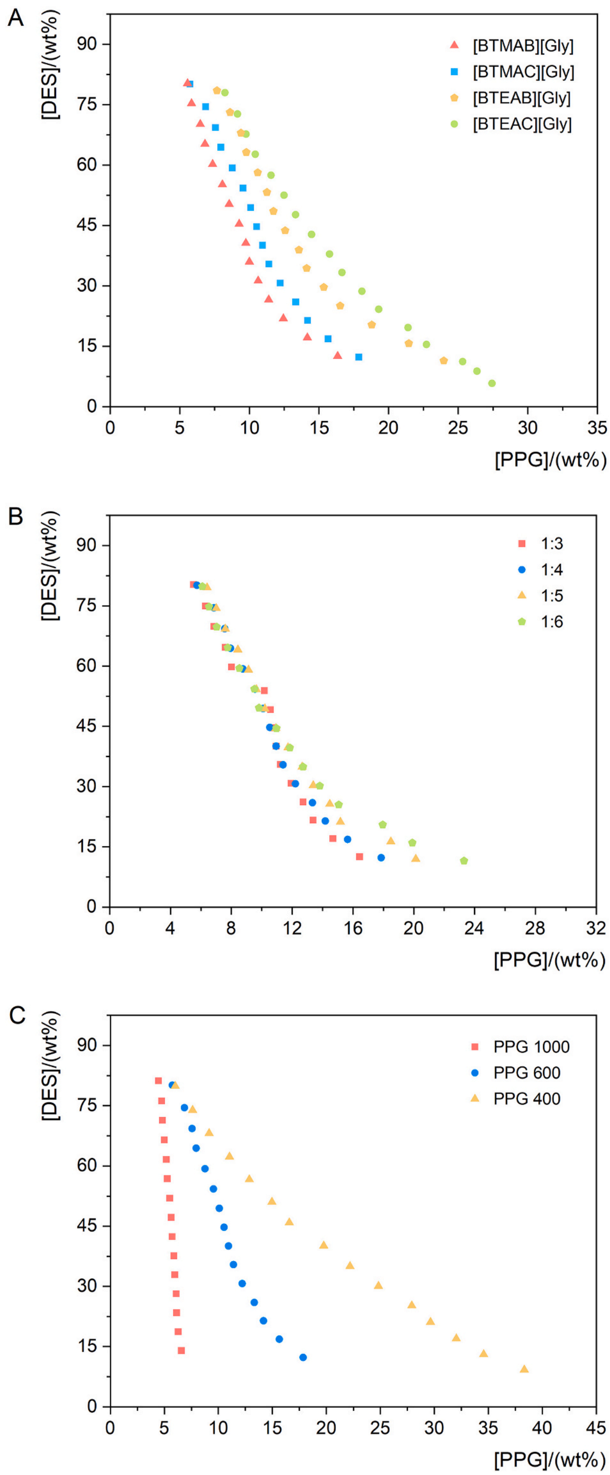

Fig. 1. Phase diagrams of DES/PPG-based ABSs: (A) effect of the types of DES, (B) effect of the molar ratio of HBA: HBD, (C) effect of the PPG molecular weights.

to the anion, it was clear that for the DESs prepared with the same quaternary ammonium salts, a change in anion from chloride to bromide led to the increase of the phase-forming ability of the DES. According to previous study, the ability of the anions to promote phase separation were related to their Gibbs energy of hydration (ΔG_{hyd}) when the ILs own the same cation (Wu et al., 2010). The more negative the ΔG_{hyd}, the stronger the salting-out capacity of the anions. As discussed before, the water structure-making components were acting as the salting-out species, thus the stronger the salting-out capacity of the water structure-making component, the stronger the phase-splitting ability. Therefore, the ILs contained Cl^- had a stronger phase-forming ability than ILs contained Br^- since the Cl^- ($\Delta G_{hyd} = -340$ kJ•mol^{-1}) had a more negative Gibbs energy of hydration than Br^- ($\Delta G_{hyd} = -315$ kJ•mol^{-1}) (Marcus, 1991). However, the opposite trend was observed in the DES-based ABSs. It is supposed that the anions of DES were restricted within hydrogen bonds, while anions of ILs had freedom to intact with others. As a result, different phase-forming behaviors were observed.

3.2.2. Effect of the molar ratio of HBA:HBD

A DES was prepared by a proper combination of HBA and HBD, thus the properties of DES were related to the molar fraction of the two components. Herein, the effect of the HBA:HBD ratio on the formation of ABSs was investigated. [BTMAC][Gly] and PPG 1000 were selected as the phase-forming components and the result was presented in Fig. 1B. The ability of the DESs with different molar ratios to form ABS followed the order: 1:3 > 1:4 > 1:5 > 1:6. As discussed before, the phase-forming ability of the component depended on its affinity for water. The log K_{ow} of BTMAC and Gly were − 2.17 and − 1.76, respectively. It was clear that the proportion of the relative hydrophobic Gly increased when the molar ratio decreased from 1:3 to 1:6, which resulted in a decrease in the hydrophilicity of DESs. In other words, the phase-forming ability of DES decreased with the decrease of the molar ratio of HBA:HBD. Since the viscosity of the [BTMAC][Gly] decreased as the molar ratio varied from 1:3 to 1:6, meanwhile, the phase-forming ability of the [BTMAC][Gly] with molar ratio of 1:4 was second only to that of [BTMAC][Gly] with molar ratio of 1:3, and there was little difference between them. Thus, the molar ratio of 1:4 was selected to continue the following experiments for its moderate viscosity and strong phase-forming ability.

3.2.3. Effect of the PPG molecular weights

PPG with different molecular weights was used to form ABS with [BTMAC][Gly] and the phase diagrams were depicted in Fig. 1C. The ability of PPG to induce phase separation obeyed the following trend: PPG 1000 > PPG 600 > PPG 400. The PPG was acted as a water structure-breaking component in the ABS, which meant its phase-forming ability was converse to its affinity towards water. The larger molecular weights of PPG rendered it more hydrophobic, and then reduced its affinity for water, thus promoting phase splitting. Apparently, the results demonstrated that the polymers with stronger hydrophobicity were more likely to form ABS.

3.3. The stability of DES in ABSs

It is reported that the integrity of DES can be destroyed in ABS due to the disruption of the hydrogen bond between the components by water (Passos, Tavares, Ferreira, Freire, & Coutinho, 2016). When the DES broke down, a nonstoichiometric partition of HBA and HBD between the coexisting phases can be observed. Thus, to explore the DES stability in the PPG-based ABSs, the ^{1}H NMR was used to characterize the top and the bottom phases to determine the molar ratio of HBA and HBD in each phase. [BTMAC][Gly] (1:4) and PPG 1000 were selected as the phase-forming agents. As illustrated in Fig. S8, after the formation of ABSs, the spectra of the PPG-rich phase were similar to that of the pure PPG, and the peaks of DES were not detected in the PPG-rich phase. This phenomenon maybe related to little DES partitioned in the top phase. As for the DES-rich phase, the molar ratio of BTMAC to Gly calculated by the integral area was 1:4, which was consistent with the original proportion. The results indicated that the molar ratio of HBA and HBD in DES was maintained in the coexisting phases of the DES/PPG-based ABSs. To further verify the stability of DES in the systems, the water contents of PPG-rich phase and DES-rich phase were determined, and the obtained values were 5.54% (wt%) and 12.32% (wt%), respectively. Hammond et al. have explored the effect of water upon ChCl-urea DES (Hammond, Bowron, & Edler, 2017). The results demonstrated that water had slight influence on the hydrogen-bonding network in DES when the water content was lower than 6.48% (wt%). DES clusters still existed when water content was between 12.28% and 40.95%. Meanwhile, it was verified that the stability of DES in water was reinforced by the increase of its hydrophobicity (van Osch, Zubeir, van den Bruinhorst, Rocha, & Kroon, 2015). The log K_{ow} of BTMAC, Gly, ChCl and urea were − 2.17, −1.76, −5.16 and − 2.11, respectively. It was considered that the [BTMAC][Gly] was more hydrophobic than ChCl-urea. Thus, it was inferred that the [BTMAC][Gly] was stable in the ABS when the water content of the DES-rich phase was 12.32%. Besides, compared with the PPG-rich phase, the DES-rich phase displayed higher water content, which also proved that the DES was the water structure-making component and the PPG acted as the water structure-breaking agent.

3.4. Construction of DES/PPG-based ABSs

Compared to AM, TA and SY, the SU was relatively hydrophobic. Since a mixture of hydrophilic and hydrophobic pigments was a better choice to explore the relative hydrophilicity-hydrophobicity of the coexisting phases in the DES/PPG and DES/salt ABSs, thus the SU was selected as the hydrophobic pigment to perform the analysis of mixed samples. The three hydrophilic pigments (AM, TA and SY) were chosen as the targets to investigate the effect of the types of DES, the molar ratio of HBA:HBD and the PPG molecular weights on the distribution of the pigments.

3.4.1. Effect of the types of DES

To evaluate the extraction performance of DES/PPG-based ABSs, PPG 1000 combined with DESs ([BTMAC][Gly], [BTMAB][Gly], [BTEAC][Gly] and [BTEAB][Gly]) was used to partition pigments (AM, TA and SY). It can be seen from Fig. 2A that the pigments were all preferentially partitioned in the DES-rich phase and the extraction efficiencies were higher than 80%, indicating the strong extraction ability of the ABSs. Besides, the extraction efficiency varied with the changes in the types of DES and pigments. Firstly, for each pigment, the extraction efficiencies of four DES-based ABSs obey the following trend: [BTMAC][Gly] > [BTEAC][Gly], and [BTMAB][Gly] > [BTEAB][Gly]. That is, the growth of the carbon chain in DES increased the steric hinderance of DES, which decreased the accessibility of the pigments for the DES. Meanwhile, the advanced viscosity of DES caused by the increasing carbon chain also prevented the pigment from entering the DES-rich phase. However, for the DESs containing the same quaternary ammonium salts, a change in anion from Cl^- to Br^- showed no clear correlation. Secondly, for each DES, the ability to extract three pigments varied with the characteristics of the pigments themselves. For [BTMAC][Gly] and [BTMAB][Gly], their extraction on pigments followed the order: AM > TA > SY. This trend may be associated with the hydrophobicity of the three pigments. The lower the log K_{ow}, the more the pigment tended to enrich in the hydrophilic phase. As illustrated in Table S1, the log K_{ow} of AM, TA and SY was −2.60 (Ferreira et al., 2016), −1.76 (Ferreira et al., 2016) and − 1.18 (Pirvu et al., 2020), respectively. Besides, the DES-rich phase and the PPG-rich phase corresponded to the hydrophilic layer and the hydrophobic layer, respectively. Thus, a decrease in log K_{ow} values was conducive to the migration of the pigments into the DES-rich phase, leading to a higher extraction efficiency of the pigments. However, for [BTEAC][Gly] and [BTEAB][Gly], although the extraction efficiency of AM was still higher than that of TA, the extraction efficiency of SY was the highest. As mentioned before, the longer carbon chain of DES rendered the DES' viscosity higher. Thus, in systems based on [BTEAC][Gly] and [BTEAB][Gly], SY (MW = 452.37) is easier to enter the DES-rich phase than AM (MW = 604.47) and TA (MW = 534.36) due to its smallest molecular weight.

3.4.2. Effect of the molar ratio of HBA:HBD

To explore the influence of the molar ratio of HBA:HBD in DES on the extraction efficiency of pigments, DESs consisting of BTMAC and Gly with different molar ratios (1:3, 1:4, 1:5 and 1:6) were prepared and utilized to form ABSs with PPG 1000. As illustrated in Fig. 2B, the extraction efficiency of all pigments nearly remained unchanged as the molar ratio ranged from 1:3 to 1:6, which indicated the molar ratio had little effect on the extraction of pigments.

3.4.3. Effect of the PPG molecular weights

The DES of [BTMAC][Gly] with a molar ratio of 1:4 and three kinds of PPG (PPG 400, PPG 600 and PPG 1000) were prepared to form ABSs. The extraction behaviors on three pigments were shown in Fig. 2C. With an increase in the PPG average molecular weight, the extraction efficiency of each pigment started to rise gradually. This was because as the molecular weight increased, the hydrophobicity of PPG increased and then the hydrophobicity of the PPG-rich phase also increased. According to the principle of similar phase dissolution, the hydrophilic pigments of AM, TA and SY tended to partition in the hydrophilic phase, thus more pigments transferred to the DES-rich phase with the increase of PPG molecular weight.

3.5. Influence factor experiments

Herein, [BTMAC][Gly] and PPG 1000 was chosen to form ABSs, and TA was chosen as the target pigment. Meanwhile, to figure out the difference between DES/polymer-based ABSs and traditional DES/salt-based ABSs, K_2HPO_4 was used as a phase-forming agent to construct DES/salt-based ABSs. The effect of several key parameters was explored in detail, namely, the mass of phase components, ionic strength and extraction time.

3.5.1. Effect of the mass of DES

The influence of DES mass on the extraction efficiency of the pigment was illustrated in Fig. 3A. It was clear in DES/PPG-based ABS, the

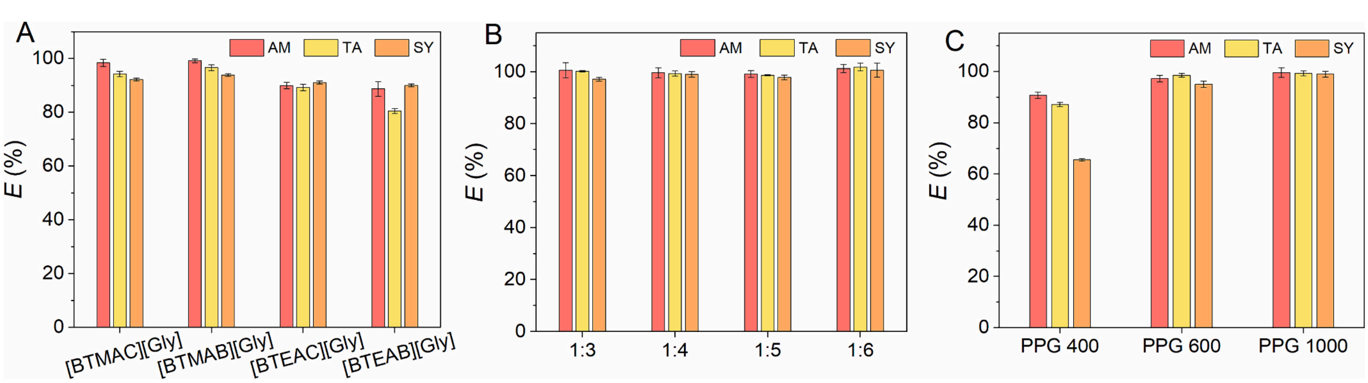

Fig. 2. Extraction results of pigments by DES/PPG-based ABSs: (A) effect of DES types, (B) effect of the BTMAC to Gly molar ratio, (C) effect of PPG molecular weights.

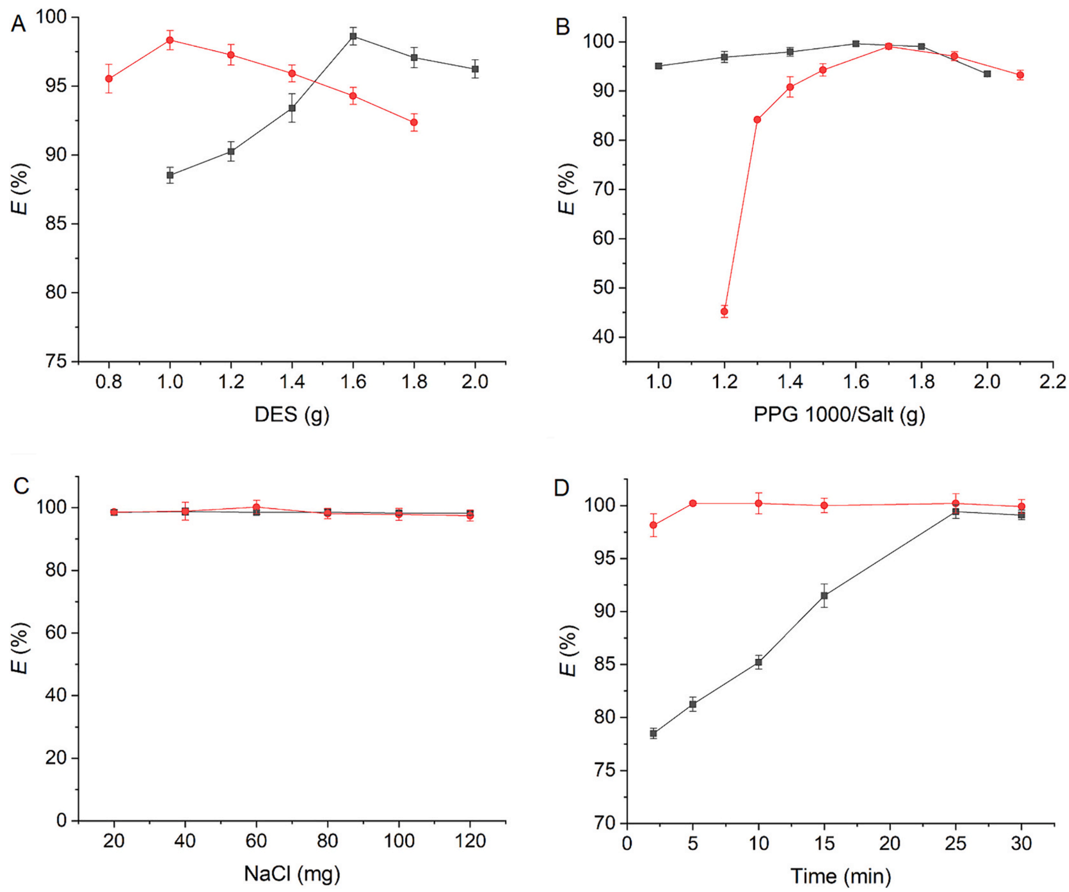

Fig. 3. Extraction of TA by DES/PPG-based ABSs (black line) and DES/salt-based ABSs (red line), respectively. (A) effect of the mass of DES, (B) effect of the mass of PPG/salt, (C) effect of the ionic strength, (D) effect of the extraction time. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

extraction efficiency increased when the mass of DES changed from 0.8 g to 1.0 g, and then started to decrease with the further increase of DES mass. This was because the bottom phase volume became larger with the continuous addition of DES, leading to more TA being transferred into the DES-rich phase. However, the viscosity of the bottom phase solution increased gradually with the increasing concentration of DES, which hindered the TA from entering the DES-rich phase. The same trend was detected in the DES/salt-based ABS. And 1.6 g and 1.0 g were selected as the optimum mass for the following experiments in the DES/PPG-based ABS and DES/salt-based ABS, respectively.

3.5.2. Effect of the mass of PPG/salt

It can be seen from Fig. 3B that the extraction efficiency increased first and then decreased with the continuous addition of PPG or salt in both DES/PPG-based ABS and DES/salt-based ABS. Taking DES/PPG-based ABS as an example, the reason was that the hydrophobicity of the top phase was increasing in the presence of relatively high PPG mass, in which the solubility of the TA in the top phase was decreased. However, the further increase of PPG mass led to a decrease of the water content of the DES-rich phase, which caused the preference of TA to transfer into the top phase. And the optimum masses were set as 1.6 g and 1.7 g for PPG 1000 and salt, respectively.

3.5.3. Effect of the ionic strength

The ionic strength always plays an important role in the distribution of analytes in ABSs. The ionic strength was related to the concentration of the salt in the solution according to the equation $I = \frac{1}{2}\sum_{i=1}^{n} c_i z_i^2$, in which I is the ionic strength of the solution, c is molal concentration of the inorganic salt ion in the aqueous solution and z represent its valence (Ricciardi, Verboom, Lange, & Huskens, 2022). Thus, the influence of the ionic strength on the pigment partition was evaluated by adding NaCl from 20 mg to 120 mg. Fig. 3C showed that the extraction efficiency of TA remained nearly unchanged in the studied range in two systems. Besides, the extraction efficiencies were also pretty much the same as the no-adding NaCl systems. The results revealed that the electrostatic interaction was not a major driving force in the separation of TA. And NaCl was not added in the following ABSs.

3.5.4. Effect of the extraction time

Different shaking time (2, 5, 10, 15, 25 and 30 min) were adopted to investigate the equilibrium time of the systems. As shown in Fig. 3D, the extraction efficiency increased rapidly in the first 25 min and then attained equilibrium in DES/PPG-based ABSs. The same tendency was detected in DES/salt-based ABSs and the corresponding time was 5 min. As a result, 25 min and 5 min were selected as the shaking time for DES/PPG and DES/salt systems, respectively.

3.6. Analysis of mixed sample

As mentioned above, the AM, TA and SY preferred to partition in the DES-rich phase in DES/PPG-based ABSs. And the TA showed the same distribution trend in DES/salt systems. Considering the AM, TA and SY are hydrophilic, SU was adopted as a kind of hydrophobic pigment to further investigate the relative hydrophilicity-hydrophobicity of the two phases in the DES/PPG and DES/salt systems. The distribution of the mixed sample of TA/SU in two systems was explored as illustrated in Fig. 4. For DES/PPG-based ABS, 98.47% of SU enriched in the top phase while 99.47% of TA was gathered in the bottom phase. However, in the

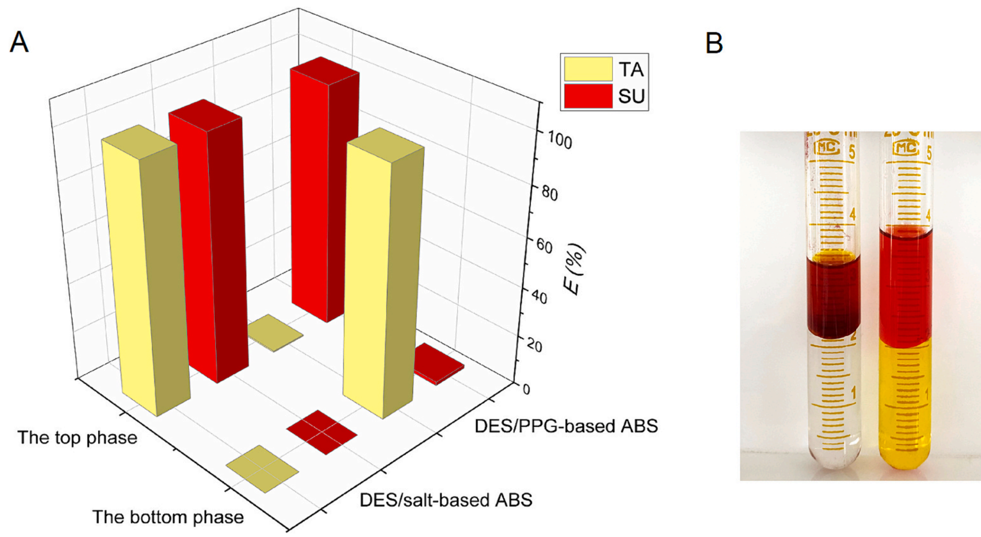

Fig. 4. (A) Extraction of mixed sample of TA/SU by DES/salt-based ABS and DES/PPG-based ABS, respectively, (B) pictures of the distribution of TA/SU in DES/salt-based ABS (the left) and in DES/PPG-based ABS (the right).

system of DES/salt, both the SU and the TA were almost completely distributed in the top phase. The reason was that in the DES/PPG system, as mentioned previously, the DES-rich phase and the PPG-rich phase were the hydrophilic layer and the hydrophobic layer, respectively. The high logK_{ow} value (7.47) of SU indicated its affinity for more hydrophobic solvents (Ferreira, Coutinho, Fernandes, & Freire, 2014), thus the SU preferred to migrate to the PPG-rich phase. As for DES/salt-based ABS, the K_2HPO_4 was an inorganic salt with a strong salting-out effect, which led to all of the pigments distributed in the DES-rich phase. To further illustrate the influence of the characterizations of the coexisting phases on the partitioning of pigments, the distribution behaviors of the mixed samples including AM/SU and SY/SU in two systems were also investigated and the results were shown in Fig. S9. Similarly, selective partitioning of hydrophilic pigments (AM and SY) and hydrophobic pigment (SU) was achieved in the DES/PPG system, while the complete distribution of the pigments in the upper phase was obtained in the DES/salt systems. The results demonstrated the distribution of the pigments was related with the hydrophilicities of both the pigments and the phase components. Besides, the results also revealed that the developed DES/PPG-based ABSs had the potential to separate pigments selectively.

3.7. Analysis of real sample

To assess the accuracy of the developed DES/PPG-based ABSs, analysis of real samples was carried out by selecting carbonated drinks, energy drinks and juice as the actual beverage samples. The measurement of the beverage samples indicated that there were no AM, TA and SY detected in the energy drinks and juice. TA was detected in the carbonated drinks and the content of TA was 3.6 μg·mL^{-1}. Afterwards, the DES/PPG-based ABSs was used to detect TA from the carbonated drinks under the optimal conditions. The obtained extraction efficiency was 98.15%, which demonstrated that the established method was able to be applied to the detection of pigments in actual beverage samples. Besides, recovery experiments were performed by spiking different concentrations of AM standards (10 and 25 μg•mL^{-1}), TA standards (5 and 10 μg•mL^{-1}) and SY standards (5 and 10 μg•mL^{-1}) to the three beverage samples. The results were summarized in Table 1. The recoveries of the three real samples ranged from 93.43% to 102.15% with the relative standard deviation (RSD) lower than 3.69%, which further indicated that the method was accurate and applicable to detect AM, TA and SY in actual samples.

3.8. Method validation

The reliability of the DES/PPG-based ABSs was assessed by the analytical characteristics containing the limit of detection (LOD) and limit of quantitation (LOQ) as well as the methodology experiments including the precision experiments, stability experiments and repeatability experiments. The LOD and LOQ were calculated as the lowest concentration of the pigment at a signal-to-noise ratio (S/N) of 3 and 10, respectively. The LODs were found 0.26, 0.24 and 0.023 μg•mL^{-1} for AM, TA and SY, respectively, and the corresponding LOQs obtained were 0.88, 0.79 and 0.078 μg•mL^{-1}, respectively. The other calculated analytical parameters were shown in Table S3. The apparatus precision was carried out by detecting the DES-rich phase for five times, and the obtained RSD were 0.43%, 0.55% and 0.28% for AM, TA and SY, respectively. Concerning stability experiments, a sample was monitored for five consecutive days, and the RSD were 0.45%, 1.58% and 0.82% for AM, TA and SY, respectively. Finally, the repeatability experiments were conducted by measuring five parallel samples, and the gained RSD were

Table 1
Determination of pigments in food samples with the proposed method ($n = 3$).

Sample	AM			TA			SY		
	Added (μg•mL^{-1})	Recovery (%)	RSD (%)	Added (μg•mL^{-1})	Recovery (%)	RSD (%)	Added (μg•mL^{-1})	Recovery (%)	RSD (%)
Carbonated drinks	10	96.26	1.09	5	95.74	1.38	5	94.78	1.86
	25	100.42	0.77	10	98.34	0.62	10	97.46	0.81
Energy drinks	10	98.10	0.79	5	97.16	1.09	5	98.46	3.69
	25	102.15	1.86	10	98.82	0.12	10	95.02	1.42
Juice	10	97.61	1.34	5	94.91	1.41	5	93.43	0.54
	25	99.84	1.48	10	99.23	0.89	10	95.23	1.04

0.84%, 1.05% and 0.78% for AM, TA and SY, respectively. The results suggested that the methods possessed good stability and repeatability.

3.9. Mechanism investigation

The exploration of the extraction mechanism is in favor of designing high-performing ABSs. Herein, the mechanism was discussed by taking the extraction of TA in the DES/PPG-based ABSs as an example.

The UV–vis and FT-IR spectrums were determined to probe into the structural changes of the DES and TA in the extraction process. The UV–vis spectra of TA before and after extraction were depicted in Fig. S10A. Clearly, the TA in the DES-rich phase after extraction displayed a similar absorption curve to that of the TA in pure water, and both of the maximum absorption peaks appeared at 427 nm. The results revealed that there was no chemical bond formed between TA and DES. Fig. S10B showed the FT-IR spectra of DES, TA and TA in the DES-rich phase. As mentioned before, for DES, the stretching vibrations at 1043 cm^{-1} and 1111 cm^{-1} belonged to the $\nu_{C\text{-}O}$. After the extraction of TA, the stretching vibrations of the C—O moved to 1109 cm^{-1} and 1037 cm^{-1}, respectively. Meanwhile, the stretching vibration of C—O in the spectra of TA also moved from 1126 cm^{-1} to 1109 cm^{-1} in the extraction process. The shift of the IR absorption peaks to lower wavenumbers indicated the formation of hydrogen bonds between DES and TA. Thus, it was inferred that hydrogen bonding may be a major force in the extraction of pigments.

Considering the chemical structures of the pigment changes with the varied solution pH, the extraction behavior of the TA under different solution pH values was investigated. The results were illustrated in Fig. S11A. It was clear that the extraction efficiency remained almost unchanged in the pH range of 2–8. Then with the further increase of the pH value, the TA tended to migrate to the top phase. The reason was that TA is a monoprotic acid which has protonated and deprotonated forms in the studied pH range (as shown in Fig. S11B). And the two forms were related to the equilibrium constant (pK_a) of the TA, which was reported as 9.5 (Pérez-Urquiza & Beltrán, 2001; Rashidi, Sajjadi, & Mousavi, 2019). When the pH is lower than 9.5, the TA mainly exists in the protonated form (HTA^{3-}). While the pH is higher than 9.5, the TA mainly exists in the deprotonated form (TA^{4-}). And the deprotonated TA stated to appear with the pH exceeded 8.0 (Sajjadi & Abdollahi, 2011). Thus when the pH was lower than 8.0, the TA was protonated with abundant -OH, it was inferred that the hydrogen bonding between TA and DES facilitated the distribution of TA in the DES-rich phase. However, with a gradual increase of pH, the protonated TA gradually transformed to the deprotonated form, leading to the weakening of the hydrogen bond force between the TA and DES. As a result, part of the TA migrated to the PPG-rich phase. The results indicated that the hydrogen bonding was a driving force in the extraction procedure. This conclusion was also confirmed by our FT-IR spectra.

To get further insight into the extraction mechanism, the size distributions of the TA solution, DES-rich phase and DES-rich phase after TA extraction were determined by the dynamic light scattering (DLS). As shown in Fig. 5, the hydrodynamic radius distributions were 108 nm, 489 nm for the TA solution and 80 nm, 311 nm for the DES-rich phase. After the extraction of TA, the hydrodynamic radius distributions of the DES-rich phase shifted to 1635 nm and 5468 nm. It was reported that the DES aggregates can be formed in the ABSs and it had good solubilizing ability for the analytes through hydrogen bonding (Deng et al., 2017; Meng et al., 2019). Thus, combined with the analysis of the FT-IR spectra and the effect of solution pH on TA extraction, it was inferred that the increase of the hydrodynamic radius after extraction was resulted from the formation of DES-TA aggregates. The TA was extracted by DES aggregates through hydrogen bonding, leading to the increase of the hydrodynamic radius of the aggregates.

3.10. Comparison with conventional solvents and other reported ABS methods

To illustrate the advantages of the developed ABSs constructed in our work, the extraction of pigments with conventional organic solvents was performed. Dichloromethane, ether, cyclopentane, carbon tetrachloride and ethyl acetate were chosen as the solvents for their water insolubility and diverse polarities. The results demonstrated that these solvents showed no extraction on AM, TA and SY, which suggested the superiority of the DES/polymer-based ABSs in pigments extraction. In addition, a comparison was made between DES/polymer-based ABSs and other reported systems in the previous literature. As shown in Table S4, five kinds of typical ABSs, including polymer/salt (Karma, Gande, & Pushpavanam, 2023), alcohol/salt (Sanglard et al., 2018), IL/salt (Yao, Gan, Li, Tan, & Shi, 2021), DES/salt (Zafarani-Moattar, Shekaari, Fattah, & Mokhtarpour, 2022) and IL/polymer (Pereira et al., 2013), were selected as the representative ABSs. Compared to the four salt-based ABSs, the DES/polymer systems exhibited higher extraction efficiencies for pigments. Besides, for the systems composed of designable solvents (IL, DES) and salt, the complete partitioning of pigments in the IL/DES-rich phase was obtained. However, the selective separation of pigments was achieved in the DES/polymer systems, which indicated the introduction of polymer into DES-based ABSs extended the hydrophilic-hydrophobic range of the coexisting phase. Additionally, although the IL/polymer-based ABS could selectively separate pigments, the complex preparation and purification of IL make the whole process

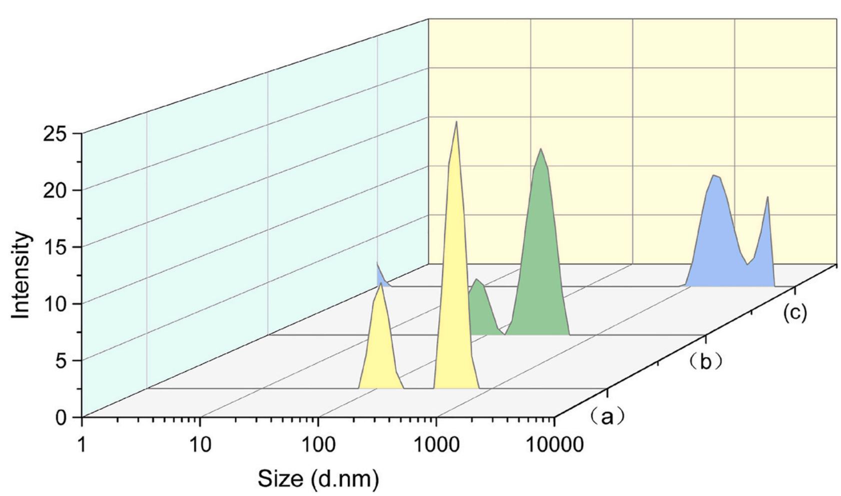

Fig. 5. Size distributions of (a) TA solution, (b) DES-rich phase, (c) DES-rich phase after extraction.

more time-consuming and complicated compared to that of DES-based ABSs. Therefore, the ABSs proposed in this study offered an alternative strategy for the efficient and selective extraction of pigments.

4. Conclusion

In summary, a novel ABS composed of DES and polymer was constructed and applied for the extraction of pigments. According to the experiment results, the more hydrophilic pigments were easier to migrate to the DES-rich phase. After the exploration of the influence factors, the extraction efficiency of TA could reach 99.27%. In comparison with the ABS formed by DES and inorganic salt, the hydrophilic-hydrophobic range was expanded in the DES/polymer system, which can be confirmed by the selective partitioning of pigments in the polymer-based system. Additionally, the DES/polymer system was successfully used to extract pigments from the actual beverage sample. Moreover, on the basis of the analysis of the extraction mechanism, the hydrogen bonding and the formation of aggregates played important roles in the extraction process. Above all, the work has extended the types of ABSs, and the versatility of the DES/polymer-based ABSs makes it a new strategy in the fields of separation and purification.

CRediT authorship contribution statement

Kaijia Xu: Writing – review & editing, Writing – original draft, Funding acquisition, Formal analysis. **Yan Zhao:** Investigation, Formal analysis. **Kai Chen:** Investigation. **Huiru Dong:** Formal analysis. **Sa Sun:** Formal analysis. **Ziyi Ni:** Investigation. **Yuzhi Wang:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foodchem.2024.139206.

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